BTS-V(R/S)-11-24-3098 Reg. No. | | | | | | | | | |

B

B.Tech. Degree V Semester Regular/Supplementary Examination November 2024

CS/EC/IT/CE/EE/ME/SE 19-200-0501 NUMERICAL AND STATISTICAL METHODS

(2019 *Scheme*)

Time: 3 Hours Maximum Marks: 60

Course Outcomes
On successful completion of the course, the students will be able to:

CO1: Solve algebraic and transcendental equations by numerical methods.
CO2: Solve numerical differentiation and integration problems.
CO3: Compute the mean and variance of probability distribution including the binomial distribution.
CO4: Test hypothesis on data.
Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

PART A
(Answer *ALL* questions)

			Marks	BL	CO	PO
		(8 × 3 = 24)				
I.	(a)	Find a positive value of $(48)^{1/3}$ correct to three decimal places using Newton - Raphson method.	3	L2	1	2
	(b)	Form the divided difference table for the following data.	3	L2	1	1

x	-4	-1	0	2	5
y	1245	33	5	9	1335

	(c)	Evaluate $\int_0^1 \frac{1}{1+x}dx$ using Trapezoidal rule by taking h = ¼.	3	L3	2	2
	(d)	Given $\frac{dy}{dx} = y - \frac{2x}{y}$, $y(0)=1$; Find y for $x = 0.2$ by Euler's method. Take h = 0.1.	3	L2	2	1
	(e)	If $f(x) = \begin{cases} \frac{1}{2}(x+1), & -1<x<1 \\ 0, & \textit{elsewhere} \end{cases}$ represents the density function of a random variable X, Find $E(X)$ and $V(X)$.	3	L5	3	1
	(f)	Suppose X has a Poisson distribution. If $P(X=2) = \frac{2}{3}P(X=1)$, then find P $(X=2)$ and P $(X \geq 2)$.	3	L2	3	3
	(g)	If a random sample of size n with sample mean $\overline{X}$ is taken from an infinite population having the mean μ and variance σ^2, then show that $E(\overline{X}) = \mu$ and $V(X) = \frac{\sigma^2}{n}$.	3	L1	4	1
	(h)	The mean weight obtained from a random sample of size 100 is 64 gms. The standard deviation of the weight distribution of the population is 3 gms. Test the statement that the mean weight of the population is 67 gms at 5% level of significance.	3	L3	4	1

(P.T.O.)

PART B

			(4 × 12 = 48)	Marks	BL	CO	PO
II.	(a)	Find the real root of the equation $x\log_{10}x=1.2$ by Regula-Falsi method correct to four decimal places.		6	L2	1	2
	(b)	Find the polynomial $f(x)$ by using Lagrange's formula and hence find $f(3)$ for		6	L3	1	2

x:	0	1	2	5
f(x):	2	3	12	147

OR

III.	(a)	Estimate the value of $f(22)$ and $f(42)$ from the following data.						6	L3	1	2

x	20	25	30	35	40	45
f(x)	354	332	291	260	231	204

	(b)	Solve the following system of equations by Gauss-Seidel method: $6x-y-z=19,\ 3x+4y+z=26,\ x+2y+6z=22.$						6	L3	1	1
IV.	(a)	The distance covered by an athlete for 50 metre is given in the following table.						6	L2	2	3

Time (Sec)	0	1	2	3	4	5	6
Distance (metre)	0	2.5	8.5	15.5	24.5	36.5	50

Determine the speed of the athlete at $t=5$ sec, correct to two decimals.

	(b)	Find by Taylor Series method the value of y at $x=0.1$ and $x=0.2$ from $\frac{dy}{dx}=x^2y-1,\ y(0)=1$.						6	L4	2	1

OR

V	(a)	The velocity v of a particle at a distance s from a point on its linear path is given by following table:						6	L2	2	3

$s(m)$	0	2.5	5.0	7.5	10.0	12.5	15	17.5	20
v (m/s)	16	19	21	22	20	17	13	11	9

Estimate the time taken by the particle to traverse the distance of 20 m.

	(b)	Apply Runge - Kutta method of fourth order to solve $\frac{dy}{dx}=\frac{y^2-x^2}{y^2+x^2}$ with $y(0)=1$ at $x=0.2$ and $x=0.4$.						6	L4	2	1

(Continued)

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			Marks	BL	CO	PO
VI.	(a)	If the Probability that an individual suffers a bad reaction from an injection is 0.001, determine the probability that out of 2000 individuals, (i) Exactly 3. (ii) More than two individual. will suffer bad reaction.	6	L5	3	1
	(b)	Derive the mean and variance of binomial distribution.	6	L1	3	2
		OR				
VII.	(a)	In an intelligence test administered on 1000 childeren, the mean was 60 and the standard deviation was 20. Assuming that the marks obtained by the children is normally distributed, find the number of children who have scored (i) Between 50 and 80 marks. (ii) Over 90 marks.	6	L3	3	3
	(b)	Fit the curve of the form $y = ab^x$ for the following data by the method of least squares.	6	L5	3	1

x	1	2	3	4	5	6
y	2.98	4.26	5.21	6.10	6.80	7.50

VIII.	(a)	Fit a Poisson distribution to the following data and test for its goodness of fit at level of significance 0.05.	6	L4	4	3

x	0	1	2	3	4
f	419	352	154	56	19

	(b)	A random sample of 50 items gives the mean 6.2 and variance 10.24. Can it be regarded as drawn from a normal population with mean 5.4 at 5% level of significance?	6	L3	4	4
		OR				
IX	(a)	In the Sampling experiment two samples composed of 6 and 11 elements and $s_1 = 3.6$; $s_2 = 2.0$. Is the difference between s_1 and s_2 significant at the 5% level	6	L4	4	3
	(b)	A random sample of size 20 from a normal population gives the sample standard deviation of 6. Test the hypothesis that the population standard deviation is 9.	6	L3	4	4

Bloom's Taxonomy Levels
L1 – 8.33%, L2 – 29.17%, L3 - 33.33%, L4 – 16.67 %, L5 – 12.5%.

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